04-29 第1篇

阅读理解（说明文）

① Have you ever wondered why leaves change color in the autumn? In summer, most tree leaves are bright green, but when autumn comes, they turn into beautiful shades of yellow, orange, and red. This is not magic; it is a fascinating scientific process.

② To understand this change, we first need to know about a chemical called chlorophyll (叶绿素). Chlorophyll is the stuff that makes leaves green. It is very important because it helps plants make their own food using sunlight, water, and air. This food-making process is called photosynthesis (光合作用). During the long, sunny days of summer, trees produce a lot of chlorophyll, so the leaves stay green.

③ However, in autumn, the days become shorter, and there is less sunlight. The weather also gets colder. Because of these changes, trees know that winter is coming. They stop making food and get ready to rest. As the trees stop making food, the green chlorophyll in the leaves slowly breaks down and disappears.

④ When the green color fades (褪去) away, we can finally see the other colors that were hiding in the leaves all along. The yellow and orange colors are caused by carotenes (胡萝卜素), the same chemical that gives carrots their orange color. These colors are always in the leaves, but in summer, there is so much green chlorophyll that it covers them up.

⑤ Some leaves also turn red or purple. These colors are made by anthocyanins (花青素). Unlike the yellow and orange colors, trees only produce extra anthocyanins in the autumn when the days are bright and the nights are cold. So, the more sunny autumn days we have, the redder the leaves will be!

注释： chemical 化学物质；chlorophyll 叶绿素；photosynthesis 光合作用；fade 褪去；carotene 胡萝卜素；anthocyanin 花青素

选择题

1. What causes the green color in tree leaves during the summer? A. Yellow carotenes. B. Red anthocyanins. C. A chemical called chlorophyll. D. Rain water and cold air.

2. What does a tree use to make its own food? A. Sunlight, water, and air. B. Yellow sand and hot fire. C. Dead leaves and animals. D. Orange carrots and sugar.

3. Why do trees stop making food in the autumn? A. Because they have too much food left from summer. B. Because it rains too much and destroys their roots. C.

Because the birds eat all their leaves. D. Because the days get shorter, there is less sunlight, and the weather gets colder.

4. Form paragraph ④, why can't we see the yellow and orange colors in the summer leaves? A. Because they are washed away by the summer rain. B. Because the green chlorophyll covers them up. C. Because the trees haven't produced them yet. D. Because they only appear in the dark at night.

5. How do leaves get their red or purple colors in the autumn? A. They are painted by birds. B. The yellow carotenes turn red after dropping on the ground. C. The trees produce extra anthocyanins during bright autumn days and cold nights. D. They absorb (吸收) the red light from the sunset.

详细解析

1. 【答案】C

【题型】 细节理解

【定位】 第②段第2句 "Chlorophyll is the stuff that makes leaves green."

【解析】 原文明言，使树叶呈现绿色的物质是一种叫叶绿素（chlorophyll）的化学物质。因此选项C正确。A为胡萝卜素决定黄橙色；B为花青素决定红紫色。故选C。

2. 【答案】A

【题型】 细节理解

【定位】 第②段第3句 "...it helps plants make their own food using sunlight, water, and air."

【解析】 本段提到，叶绿素通过利用阳光、水和空气（sunlight, water, and air）来帮助植物制造自己的食物。这就直接印证了A选项。B、C、D均不符合植物光合作用的过程。故选A。

3. 【答案】D

【题型】 细节理解

【定位】 第③段 "...in autumn, the days become shorter, and there is less sunlight. The weather also gets colder... They stop making food and get ready to rest."

【解析】 树木到了秋天为什么停止制造养分呢？因为秋天白天变短，阳光减少，并且天气变冷，树木察觉到冬天即将来临所以准备休眠。这与D项（由于白天变短、阳光变少且天气变冷）完全吻合。A（食物太多）、B（雨水毁根）、C（鸟吃叶子）都是臆造的。故选D。

4. 【答案】B

【题型】 细节理解

【定位】 第④段最后一句 "These colors are always in the leaves, but in summer, there is so much green chlorophyll that it covers them up."

【解析】 夏天为什么看不到树叶里的黄橙色？原文解释说，这些颜色一直都在叶子里，但夏天大量的绿色叶绿素把它们给掩盖住了（it covers them up）。由此选择B（由于绿色叶绿素掩盖了它们）。故选B。

5. 【答案】C

【题型】 细节理解

【定位】 第⑤段第2-3句 "These colors are made by anthocyanins (花青素)... trees only produce extra anthocyanins in the autumn when the days are bright and the nights are cold."

【解析】 关于树叶如何变红变紫，第五段指出这是由花青素决定的，且树木只在秋天白天明亮且夜间寒冷时才会产生额外的花青素。这对应了C选项（树木在明亮的秋日和寒冷的夜晚产生额外的花青素）。A（鸟涂的）、B（掉地上变红的）、D（吸收落日红光）皆不符科学常识与原文表达。故选C。